must share with them certain observable features, in particular an abundance of cells with many independent inputs and low thresholds. (Brindley 1969: 173)

Various experimentally testable predictions about synapses and cell numbers could be generated accordingly. In other words, this computational approach not only 'made sense' of the brain, but promised to help illuminate its detailed structure.

Among those whom Brindley acknowledged at the end of his 1969 paper were Barlow (already widely respected) and two unknowns: his students Stephen Blomfield and Marr. As we'll see in Section v.c, Marr was soon to publish the first formal theory of the brain *as such*. Eventually, because of his later research (on vision), he would become a household name for cognitive scientists in general. Brindley, by contrast, remained largely unknown.

But his work had been crucial. It confirmed what McCulloch, Uttley, Barlow, and Gregory had already argued: that *traditional neuro-anatomy must be combined with computational analysis*, if the nervous system is to be understood.

14.iv. A Fistful of Feature-Detectors

If Barlow's theory of perceptual coding was important for the rise of computational neuroscience, so was his practical work. Indeed, some of his early experiments led indirectly to perhaps the most famous paper (and certainly the best title) in the field.

Neurophysiologists had known since the late 1930s, thanks to Keffer Hartline at Rockefeller University, that some retinal cells in frogs aren't mere passive receptors of points of light. Rather, they respond to simple patterns of illumination: ON, OFF, or ON–OFF (Hartline 1938). In the early 1950s, Barlow reported in the *Journal of Physiology* that such cells are more sensitive to edges (light contours) than to unstructured light, and that the ON–OFF variety are especially sensitive to movement (Barlow 1953). Half-jokingly, he even called these cells "bug detectors".

In 1959 that term would start to ring still-resounding bells.

a. Bug-detectors

Most people today associate bug-detectors not with Barlow, but with the four authors of 'What the Frog's Eye Tells the Frog's Brain': Lettvin, Humberto Maturana, McCulloch, and Pitts (1959).

McCulloch had been a founder of the cybernetics movement (4.v–vi), and the 'Frog's Eye' work was done at the hub of cybernetics: MIT. So it's prima facie not surprising that although three of the four authors were neuroscientists, the paper appeared—like many 'cognitive' pieces at that time—in a journal officially aimed not at biologists but at radio engineers. However, this didn't happen by choice. Lettvin remarked in a lecture, years later, that they'd tried to get the paper published in several biological journals but were "laughed off the stage". The reason was that they saw coding not as all-or-none but as implemented by variable *bursts* of nerve signals (John Collier, personal communication).

The paper's message was as memorable as its title. For it showed that the frog's eye has a great deal to tell its brain, over and above the presence of light—or even of Barlow's edges and movement.

As Lettvin's group put it: "the eye speaks to the brain in a language already highly organized and interpreted, instead of transmitting some more or less accurate copy of the distribution of light on the receptors" (Lettvin *et al.* 1959: 251). Specifically, they had found single cells in the frog's retina that were sensitive to one of four different light patterns:

> (1) local sharp edges and contrast; (2) the curvature [within specific limits] of edge of a dark object; (3) the movement of edges; and (4) the local dimmings produced by movement or rapid general darkening. (p. 253)

These results were observed even when the stimulus, instead of being artificially isolated (as in Barlow's experiments), was surrounded by a *naturalistic* image of the countryside as presented to a frog.

The 'Frog's Eye' authors also found that the four cell-types mapped onto four distinct cell sheets in the frog's brain (the optic tectum, or colliculus). And these were precisely co-registered: neighbouring points in the retina were connected to neighbouring points in sheet 1 . . . and sheet 4. What's more, the nerve fibres would grow back to the right place if they were cut. The idea of *some sort of* point-to-point mapping had been suggested by Julia Apter, and by Pitts and McCulloch in their 1947 paper (see 12.i.c). Now, there was detailed experimental evidence. (The explanation for *how* this mapping is established was then unknown: see ix.a, below.) And there were various subtleties. For instance, some class 3 cells fired only if the movement was in a certain, very broadly defined, direction.

In fact, there were some subtleties so surprising that they weren't even mentioned in the published paper. Lettvin recalls:

> What we did not report, and what to me is still [in 1994] the most astonishing thing about the bug detectors, is the following property . . . You bring one spot in, move it into the [receptive] field, and as long as you move it around, wonderful response.
>
> You bring in two spots; if they're rigidly coupled in their motion by a fixed distance between them and are moved around, you get a good response almost as if they are only one spot.
>
> You bring three rigidly coupled spots in, and it doesn't matter their size or their disposition or their distances from each other: move them around as a rigidly coupled triad and there's no response at all.
>
> . . . If you have a white background and two black spots, OK; three spots, forget it. You connect any two spots of the three by a barely visible black line, and all of a sudden it's a two-spot system. It becomes visible. Or else you move any one spot with respect to the other two, now there's a response. (J. A. Anderson and Rosenfeld 1998: 18)

How, Lettvin asked himself, could a mere retina possibly compute such distinctions? He still has no answer—so still hasn't officially published the finding. In his judgement, then, this discovery wasn't really a discovery (see Chapter 1.iii.f). Putting it in Mayhew's terms, he'd discovered the cells, and what they were computing—but not *how* they were computing it.

What's more, it seemed that the eye might be telling the brain very interesting things—interesting, that is, *to the frog*. These concerned the approach of predators (casting a shadow over the scene: see class 4 above), and the whereabouts of food:

The operations thus have much more the flavor of perception than of sensation if that distinction has any meaning now. That is to say that the language in which they are best described is the language of complex abstractions from the visual image. We have been tempted, for example, to call the convexity detectors "bug perceivers". Such a fiber responds best when a dark object, smaller than a receptive field, enters that field, stops, and moves about intermittently thereafter. The response is not affected if the lighting changes or if the background (say a picture of grass and flowers) is moving, and is not there if only the background, moving or still, is in the field. Could one better describe a system for detecting an accessible bug? (Lettvin *et al.* 1959: 253–4)

Indeed, the retinal cells seemed to guarantee the frog a healthy diet of fresh food. They responded only to stimuli that were moving, so—in its normal environment—probably *alive*. This tallied with previous observations of the whole animal's behaviour:

The frog does not seem to see or, at any rate, is not concerned with the detail of stationary parts of the world around him. He will starve to death surrounded by food if it is not moving. His choice of food is determined only by size and movement. He will leap to capture any object the size of an insect or worm providing it moves like one. He can be fooled easily not only by a bit of dangled meat but by any moving small object. (p. 231)

Remarkably, the radius of curvature that strongly triggered the convexity detectors corresponded to bug size. In short, the frog's eye was seeing only what the frog needed to know.

Later, Lettvin would complain about the use of such "anthropomorphic" language even in talk about human brains, adding that "people have screwed up the frog because they're taking bachtriomorphic views of the frog. If you really want to understand the frog, you must learn to be objective and scientific" (J. A. Anderson and Rosenfeld 1998: 344). The 'Frog's Eye' paper itself, however, had initiated this perspective.

The 'Newtonian' assumption inherited from the associationists, and dating back to John Locke's distinction between "sensation" and "reflection" (Chapter 2.x.a), had been that the sense organs—and by extension, the peripheral parts of the brain—are both passive and atomistic. Sensitivity to points of light, *yes*; to complex patterns of light, *no*. To perceive (infer, compute) a light–dark gradient with a specific radius of curvature, it had been tacitly assumed, was a matter for the cerebral cortex—present only in mammals and birds—and must involve a large number of cells. Now, that assumption was shattered.

But the surprise, though great, wasn't unqualified. Recent work in both neurophysiology and cybernetics/AI had implied that some such results might be found. Barlow's experiments a few years before had modified the assumption already. Indeed, others besides the MIT team were looking for feature-detectors of some sort (see below). And Pitts and McCulloch (1947) themselves had posited several layers of visual computation in the brain, going from simple features to increasingly complex concepts.

(Admittedly, they'd assumed that the features would be relatively simple, and formally tractable. Pitts was therefore, so Lettvin recalls, "a little appalled by the results, which were very different from what he expected", and "while he accepted the work enthusiastically, at the same time it disappointed him"—J. A. Anderson and Rosenfeld 1998: 10.)

Moreover, many cyberneticians—though not the orthodox neurophysiologists—had encouraged attention to what *the animal* needs to know, given what it wants to do. In particular, Lettvin's mentor McCulloch had taught him to think of sensory neurophysiology as "experimental epistemology". (I remember being startled to find these words written on the door, on my first visit to Lettvin's lab.) He meant that one should ask not merely how perception tells animals about the world, but also how it makes appropriate action possible (Chapter 4.iii.b).

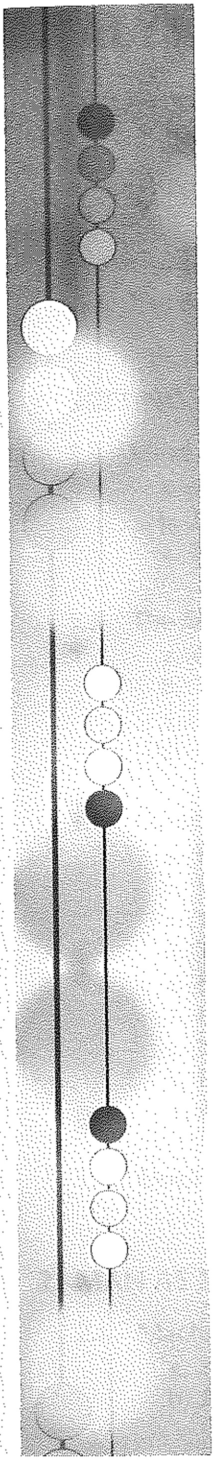
A visual neurophysiologist who became close to the MIT team a few years later was told by Lettvin that "Pitts and McCulloch actually were on the eye/brain paper as a friendly courtesy. They had no role in the planning or execution or anything of the eye/brain thing" (C. G. Gross, personal communication). However, McCulloch's lack of hands-on contributions doesn't rule out his having had a background influence on Lettvin's thinking.

Similarly, Selfridge had argued—on computational grounds, and partly influenced by Pitts and McCulloch—that complex perceptions *must* be based on many distinct feature-detectors (or "data demons"), where the "features" have survival value for the animal. And he'd illustrated part of what he meant by the Pandemonium program. (Only "part", because the program didn't need to survive, so didn't *want* anything: see Chapter 12.ii.d and iv.b.)

Lettvin was a very close friend of Selfridge, and of McCulloch and Pitts. Their mutual friendship had led to the 'Frog's Eye' work in the first place. The paper described itself as "an outgrowth" of the 1947 study of universals, which—in effect—had postulated layers of feature-detectors. In addition, it carried this acknowledgement: "We are particularly grateful to O. G. Selfridge, whose experiments with mechanical recognizers of pattern helped drive us to this work and whose criticism in part shaped its course" (Lettvin *et al.* 1959: 253, 254). In short, although the authors were doing "wet" neuroscience, not describing a computer model, they were plainly inspired by computational ideas.

What actually happened, according to Maturana (personal communication), began with Maturana's accidental discovery (in November 1958) of movement-sensitive cells in the frog's retina. He found these interesting because they confirmed his hunch that such cells might exist, a hunch based on purely anatomical grounds (microscopic evidence of asymmetries in the connection patterns of some of the retinal cells). But Lettvin's familiarity with Selfridge's ideas enabled him to see the wider importance of Maturana's finding. He immediately switched the laboratory's main research programme to a hunt for other specialized, and ecologically significant, cells—and they found them.

A further surprise was in store. Within a few years, Maturana rejected the computational interpretation of these experiments. He developed their whole-animal emphasis into a neo-Kantian philosophy of biology, in which the concepts of computation and information had no place (see 15.vii.b and 16.x.c). Orthodox neuroscientists, however, knew little or nothing of this—and cared less. Even with Maturana's blessing, neo-Kantianism was out of fashion.



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